Magnetic Dipoles: A current loop, of area A, behaves like a magnet and has a magnetic dipole moment:

$$\boldsymbol{M} = IA\hat{\boldsymbol{n}}$$

Where $\hat{\mathbf{n}}$ is a unit vector normal to the plane of the loop. Its direction is found by placing the **fingers** of the right hand along the **current**; the **right thumb** then indicates the direction of $\hat{\mathbf{n}}$ and $\mathbf{M}$. For N-current loops

$$\boldsymbol{M} = NIA\hat{\boldsymbol{n}}\ .$$

In a **uniform magnetic field**, a magnetic dipole behaves much like an electric dipole in uniform electric field, and the net force on the dipole will be zero. The torque, $\boldsymbol{\tau}$ on the dipole and the potential energy, U, are given by:

$$\tau = \boldsymbol{M} \times \boldsymbol{B}, \qquad U = MB\cos\theta$$

The magnetic torque on a flat current-carrying loop of wire by a uniform magnetic field **B** is maximum when the plane of the loop is parallel to **B**.

Example 9: The plane of area 4.0 cm^2 rectangular loop of wire is parallel to a 2.0 T magnetic field. The loop carries a current of 6.0 A. Calculate the magnitude of the torque acts on the loop.

Answer:

$$\tau = \boldsymbol{M} \times \boldsymbol{B} = NIA \times \boldsymbol{B} = NIAB\sin\theta$$

$$= (1)(6)(4\times10^{-4})(2)\sin 90^\circ = 4.8\times10^{-3}\ \text{N.m}$$

Example 11: A 100-turn circular coil of wire with radius 1.0 cm carries a current of 0.5 A. What torque will be exerted on the coil when it is placed in a magnetic field of 5.0 mT which makes an angle of 60° with the plane of the coil?

Answer:

$$\boldsymbol{M} = NIA = (100)(0.5)\pi(1.0\times10^{-2})^2\ \hat{\mathbf{k}}$$

$$= 0.0157\ \hat{\mathbf{k}}.$$

$$\tau = \boldsymbol{M} \times \boldsymbol{B} = 0.0157\times5.0\times10^{-3}\sin 30^\circ = 3.93\times10^{-5}\ \text{N.m}$$

Example 14: A current of 16 mA is maintained in a single circular loop of radius 0.32 m. An external magnetic field of 0.8 T is directed parallel to the plane of the loop.

a- Calculate the magnetic moment of the current loop.

Answer:

$$|\boldsymbol{M}| = IA = 16\times10^{-3}\times\pi(0.32)^2$$

$$= \underline{5.1\times10^{-3}\ \text{C}\cdot\text{m}^2}$$

b- What is the magnitude of the torque exerted on the loop by the magnetic field?

$$|\tau| = |\boldsymbol{M}\times\boldsymbol{B}| = |\boldsymbol{M}||\boldsymbol{B}|\sin 90^\circ = 5.1\times10^{-3}\times0.8$$

$$= \underline{4.1\times10^{-3}\text{C}\cdot\text{m}^2\text{T}}$$

Sources of Magnetic Field

1. Magnetic field of a moving charge:

To find the magnetic field B of a point charge q moving with constant velocity v at the point P, let's call the location of the moving charge at a given instant the **source point** and the point P where we want to find the field the **field point**. Experiments show that the magnitude of B is proportional to q, the particle velocity v, 1 /r^2 and to the sinφ . But the direction of B is perpendicular to the plane containing the line from source point to field point and the particle velocity v.

$$B = \frac{\mu_0}{4\pi}\frac{|q|\, v\sin\varphi}{r^2} \qquad \vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$$

Where $\mu_0/4\pi$ is proportionality constant, B has the greatest magnitude when sin φ = 1 or φ = 90 . If the charge q is negative, the direction of B is opposite. The unit of B is tesla (T).

2. Magnetic Field of a Current

The principle of superposition of magnetic fields: The total magnetic field caused by several moving charges is the vector sum of the fields caused by the individual charges.

We begin by calculating the magnetic field caused by a short segment dl of a current carrying a conductor, the total moving charge dQ in the segment is:

$$dQ = nqAdl$$

Where n is the number of the moving charged particles per unit volume, A is the cross sectional area, and Adl is the volume.

The magnitude of the resulting field dB at any point P is

$$dB = \frac{\mu_0}{4\pi}\frac{|dQ|\, v_d \sin\varphi}{r^2} = \frac{\mu_0}{4\pi}\frac{|dq|\, n v_d A dl \sin\varphi}{r^2},$$

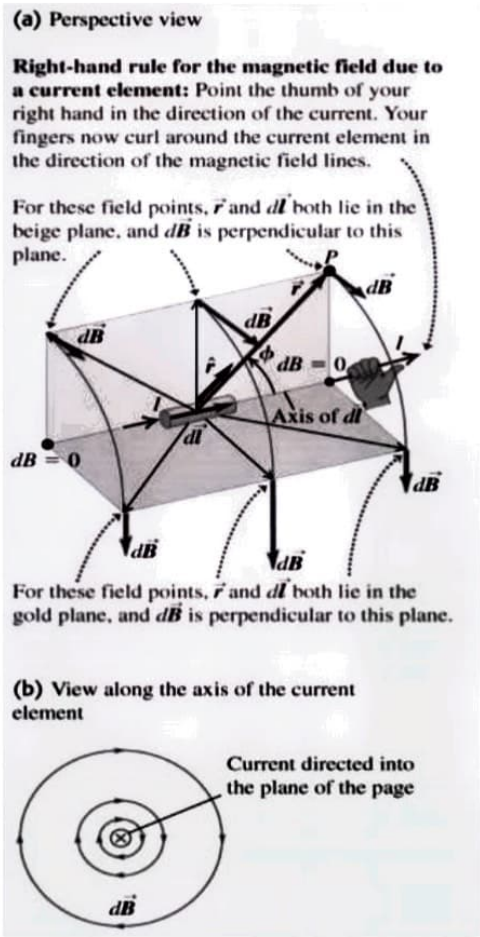
but $$n|dq|\, v_d A = I,$$

Then $$dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\varphi}{r^2}$$

In the vector form, using the unit vector r^, we have

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{l}\times\hat{r}}{r^2} \qquad \vec{B} = \frac{\mu_0}{4\pi}\int\frac{Id\vec{l}\times\hat{r}}{r^2}$$

This is the **Biot and Savart law**.



We can use this law to find the total magnetic field at any point in space due to the current in a complete circuit.

4. Force Between Parallel Conductor The lower conductor produces a B field that, at the position of the upper conductor, has magnitude:

$$B = \frac{\mu_0 I}{2\pi r}$$

The force that this field exerts on a length L of the upper conductor is:

$$\vec{F} = I'\vec{L} \times \vec{B}$$

Where the vector L is in the direction of the current I ' and has the magnitude L. Since B is perpendicular to the length of the conductor and hence to L , the magnitude of this force is:

$$F = I'LB = \frac{\mu_0 II'L}{2\pi r}$$

And the force per unit length is:

$$\frac{F}{L} = \frac{\mu_0 II'}{2\pi r}$$

Applying the right-hand rule to $\vec{F} = I'\vec{L} \times \vec{B}$ shows that the force on the upper conductor is directed downward.

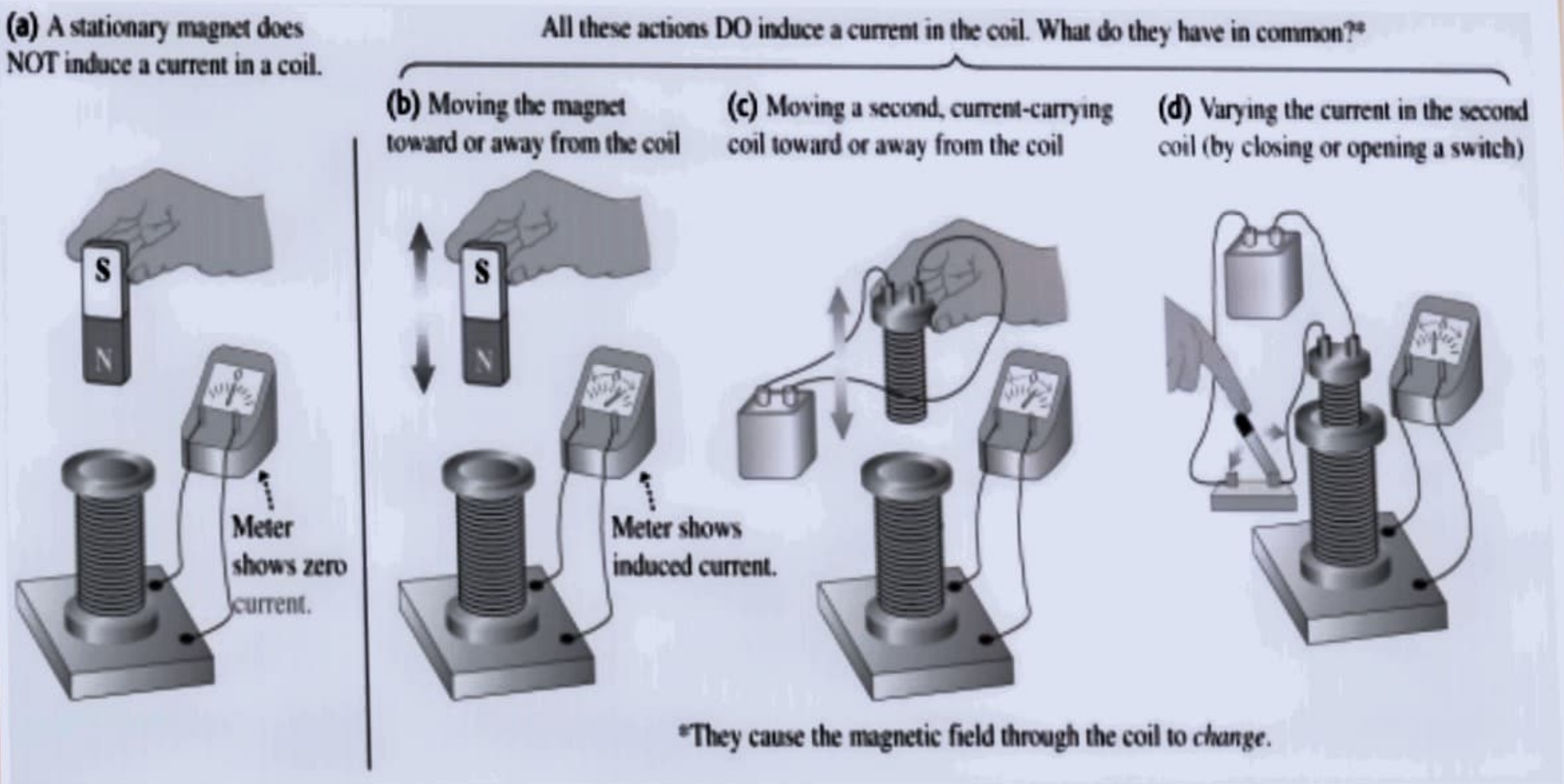
Note:

1- The two conductors carrying current in the same direction attract each other. If the direction of either current is reversed, the forces also reverse.

2- Parallel conductors carrying currents in opposite directions repel each other.

Electromagnetic Induction

1. Electromagnetic Experiments



This shows that the induced emf's that are causing the current do not depend on the material of the coil but only on its shape and the magnetic field.

Note:

The common element in all these experiments is changing magnetic flux through the coil connected to the galvanometer.